



P1 HEALTHCARE

Data integrity, access control, and patient-centered care

- Secure electronic health records
- Role-based and least-privilege access
- Consent management and revocation
- Audit logging and traceability

Healthcare × Blockchain

PBFT consensus enabling smart-contract consent automation



P2 BLOCKCHAIN

Decentralized trust, transparency, and tamper resistance

- PBFT consensus for high reliability
- Smart contracts for automation
- Immutability and non-repudiation
- Distributed ledger and auditability

BCHAIN-ACCESS FRAMEWORK

Accessibility-by-Design Blockchain Healthcare

AAD

Accessibility-by-Design

Inclusive by default

PTD

Privacy-by-Design

Data min. & user control

CID

Consent & Identity

Transparent and verifiable

UAC

Usability & Accessib.

Usable for all (PwDs)

4-way convergence (CID + AAD): biometric key binding + time-locked pending state

Secure | Accessible | Compliant | Transparent | User-Centered

Blockchain × Governance

GDPR-compliant erasure workaround using encrypted off-chain storage and pointers



P4 ACCESSIBILITY

HCI principles, inclusive design, and barrier removal

- WCAG 2.1 Level AA compliance
- Nielsen's 10 usability heuristics
- Inclusive, perceivable, operable UI
- Assistive technology compatibility

Accessibility × Governance

WCAG 2.1 as legal mandate and accessibility law alignment



P3 GOVERNANCE

Policy compliance, risk management, and ethical oversight

- GDPR, HIPAA, and local regulations
- Data protection and privacy by law
- Risk management and accountability
- Accessibility law and inclusion policies

ACCESSIBILITY FOUNDATION

WCAG 2.1 Level AA • Nielsen's Heuristics

Guiding design, development, and evaluation

PRIMARY TARGET USERS: PEOPLE WITH DISABILITIES (PwDs)



Secondary: Older adults, caregivers, and general patients

Framework refined through expert evaluation